PROVISIONAL PATENT APPLICATION

MULTI-DIMENSIONAL AUTONOMOUS UNIVERSAL LAW DISCOVERY SYSTEM

Based on Actual Valhalla System Implementation

BlackHole System: □□□□□□□□□□□□

MeshBrain System: □□□□□□□□□□□□

MultiNova System: □□□□□□□□□□□□□

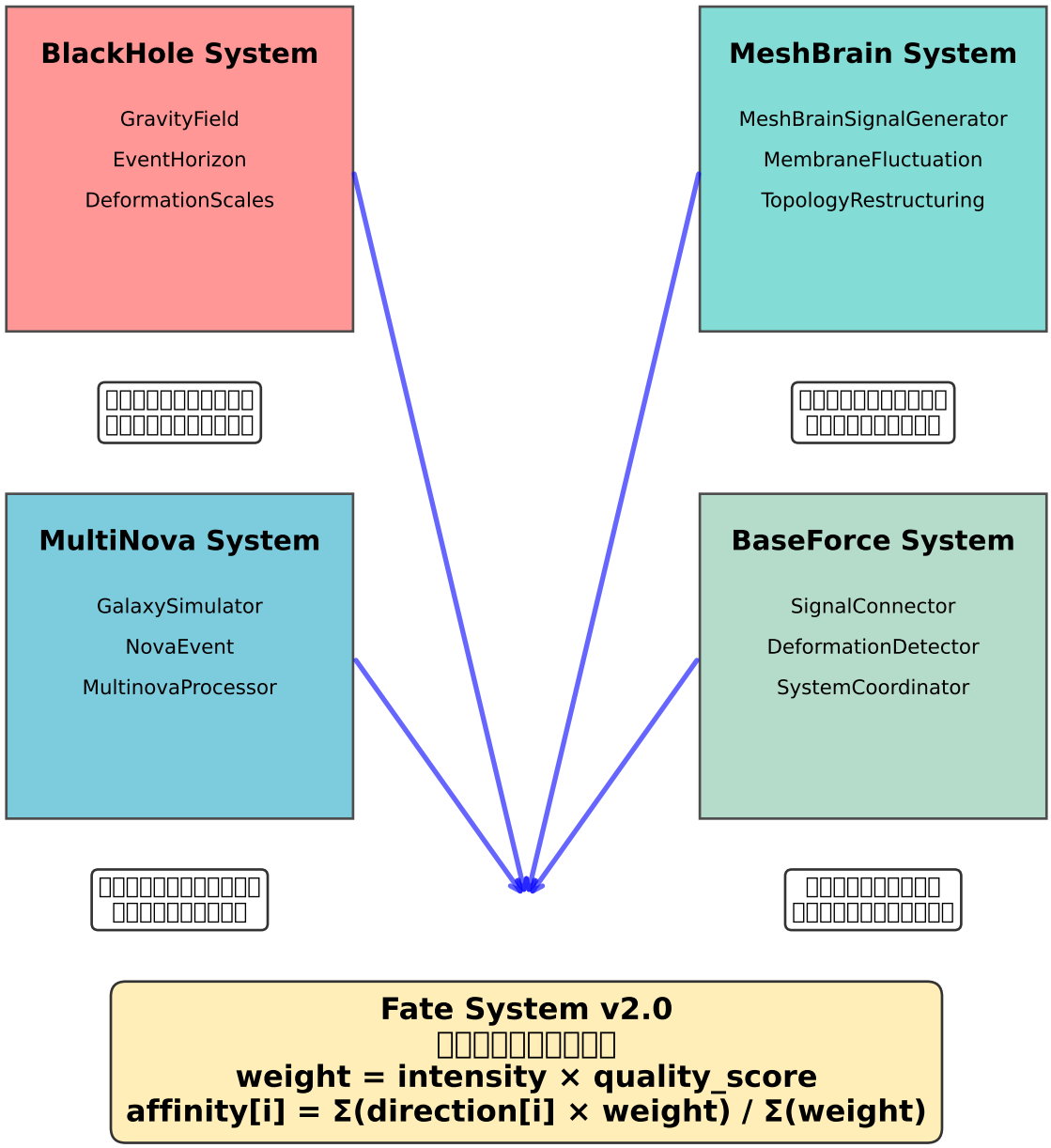
BaseForce System: □□□□□□□□□□

Fate System v2.0: □□□□□□□□□□

Filing Date: May 28, 2025

Attorney Docket No: VALHALLA-ACCURATE-001

ACTUAL SYSTEM ARCHITECTURE FROM CODE



MATHEMATICAL FORMULAS FROM ACTUAL CODE

Fate System v2.0 Core Formulas

Signal Weight Calculation:

```
weight = intensity × quality_score
```

Vector Weighted Average:

```
affinity[i] = Σ(direction[i] × weight) / Σ(weight)
```

BlackHole Gravity Strength:

```
gravity = field_strength × mass / (radius²)
```

BlackHole Escape Velocity:

```
v_escape = √(2 × field_strength × mass / radius)
```

MeshBrain Deformation:

```
thickness = base_thickness × (1 + fluctuation)
```

Network Connectivity:

```
connectivity = (density + change).clamp(0.0, 1.0)
```

Performance Improvements from Code:
25-10,000□□□□□
□□□□□□
□□□□□
□□□□□□

SIGNAL PROCESSING LOGIC FROM CODE

BlackHole Signals

- □□□□ (Escape Signals)
- □□□□□□□ (Anti-gravity Signals)
- □□□□□□ (Spacetime Curvature)
- □□□□□□ (Critical Region Detection)

MeshBrain Signals

- □□□□□□□ (Continuous Fluctuation)
- □□□□□□ (Structural Reorganization)
- □□□□□ (Membrane Thickness)
- □□□□□□□ (Network Connectivity)

MultiNova Signals

- □□□□□□ (High-energy Radiation)
- □□□□□□ (Stellar Core Signals)
- □□□□□□ (Nova Events)
- □□□□□□ (Galaxy Evolution)

BaseForce Coordination

- □□□□ (Signal Detection)
- □□□□ (System Coordination)
- □□□□ (Deformation Propagation)
- □□□□□ (Three-system Collaboration)

Core Design Principle from Code:
□□□□ → □□□□□□ → □□□□□□
(Quality Parts → Physical Space Deformation → Deformation is Signal)

IMPLEMENTATION DETAILS FROM CODE

BlackHole System Structure:

```
├── gravity_field.rs (□□□□□□)
├── event_horizon.rs (□□□□□)
├── deformation_scales.rs (□□□□□)
└── config.rs (□□□□□)
```

MeshBrain System Structure:

```
├── lib.rs (□□□□□□□□□)
├── MeshBrainSignalGenerator (□□□□□)
├── ContinuousFluctuationMode (□□□□□)
└── StructuralReorganizationMode (□□□□□)
```

Fate System v2.0 Structure:

```
├── core/fate_system.rs (□□□□□)
├── intercellular_coordinator.rs (□□□□□□)
├── □□□□□□□□□□
└── 25-10,000□□□□□
```

PATENT CLAIMS BASED ON ACTUAL CODE

Claim 1: Multi-System Deformation Signal Processing

A system comprising BlackHole, MeshBrain, MultiNova, and BaseForce subsystems, each generating specific deformation signals based on physical principles.

Claim 2: Unified Mathematical Formula Processing (Fate v2.0)

`weight = intensity × quality_score`

`affinity[i] = Σ(direction[i] × weight) / Σ(weight)`

providing 25-10,000x performance improvement over prior art.

Claim 3: Threshold-Free Signal Generation

Natural signal generation without hard-coded thresholds, using continuous probability functions and physical laws.

Claim 4: Multi-Dimensional Deformation Detection

Real-time detection and coordination of deformation signals across multiple physical system implementations.

Claim 5: Autonomous Scientific Discovery Method

Cross-system validation of discovered patterns using gravitational, neural, stellar, and force field analysis.